Rajesh Kumar Kona

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Profile

Software engineer with two years at Accenture shipping enterprise Salesforce systems under strict SLAs, promoted ahead of the standard cycle, now studying for an MSc in Computer Science at the University of Edinburgh and serving on the CompSoc committee as Masters Representative. I build complete systems end to end, put them in front of real users and test my own claims hard: an AI web app I run alone with 100,000+ pageviews, a GPU video compositor verified against a CPU reference, an agent runtime that keeps the LLM out of the safety decision, a job-search system that never submits on anyone's behalf, and a trading research lab that reports openly when nothing works. Competed in the UK & Ireland Programming Contest (UKIEPC 2026), the ICPC subregional, against the strongest university teams in the UK and Ireland. Graduated 2nd of 66 in my BTech cohort; three journal papers and a systems preprint.

Education

University of Edinburgh, Edinburgh, UK **September 2026 - Expected 2027**
MSc in Computer Science
Modules: Machine Learning Systems, Machine Learning Practical, Text Technologies for Data Science, Human-Computer Interaction, Blockchains and Distributed Ledgers, Applied Cloud Programming, Informatics Project Proposal and a 60-credit dissertation (here)
Group projects: building a search engine with retrieval-augmented generation (TTDS); designing and training our own neural network as a team (MLP); a user study observing how developers build and understand software with an AI coding assistant (HCI)
Leadership: Committee Member, CompSoc (Edinburgh University Computing Society), elected as Masters Representative

Vasireddy Venkatadri Institute of Technology, Guntur, India **September 2020 - July 2024**
Bachelor of Technology in Computer Science and Engineering (Specialisation: IoT and Cyber Security, including Blockchain)
GPA: 8.65/10 · Ranked 2nd of 66 in the cohort
Honours/Awards: three journal papers as an undergraduate (IJCRT 2024, IRJET 2023 ×2), Third Prize in the 24-Hour Design Venture Challenge (2022), project presenter at IEEE Soft-con Expo, IEEE IoT Expo and the Innovation Fair, JNTU Kakinada
Coursework: data structures, algorithms, operating systems, computer networks, DBMS, machine learning, big data analytics, cryptography and network security, cloud computing, blockchain

Experience

Accenture, Bengaluru, India **August 2024 - September 2026**
Software Engineer (Salesforce Developer), promoted from Associate Software Engineer

- Promoted to Software Engineer in about 21 months, ahead of the standard cycle, for sustained delivery, ownership and client trust
- Delivered 170+ production contributions and resolved 20+ critical defects within strict SLA timelines on enterprise CRM systems
- Caught a critical defect before release, traced it to the root cause and fixed it, preventing a significant business loss for the client; formally recognised by the onshore team (Client Recognition, 2026)
- Designed and built CRM solutions in Apex, Lightning Web Components, SOQL and Flow, automating workflows and cutting manual processing
- Built secure REST API integrations for reliable two-way data exchange with third-party enterprise services, and reusable components aligned to enterprise architecture standards
- Received the Cheer Award (2025) for performance and cross-team collaboration

Tech Stalwart Solution, India **June - July 2023**
Full-Stack Engineer Intern

- Improved page-load speed by roughly 15% with React.js and co-deployed the company website on AWS

Open-Source & Research Projects

LexoraAI, AI document workspace **2025 - Present**
Creator, built and run single-handedly (live here)

- Served 100,000+ pageviews and 6.7M+ requests in production (verified in Cloudflare analytics), on hosting and analytics I set up and run myself, behind Cloudflare
- Designed and built the whole live AI web app on my own (front end, back end, LLM integration): read-aloud with highlighting, voice commands, an AI explainer and 40+ document tools; free tools run in the browser, and server features ask consent and delete files afterwards

Voice-to-Video, open-source narration-to-video pipeline **2026 - Present**
Creator and sole engineer (code here, research preprint here)

- Built a Python system (173 source files, 63,500 lines, 1,708 test functions) that turns speech, a script or a document into a narrated, captioned video

- Profiled before optimising: composing frames took 77% of render time, so by Amdahl's law a hardware encoder could give at most 1.3×; removing redundant full-frame buffer copies instead doubled composition throughput (12 → 26 frames/s)
- Designed frame-exact 12-second segmented rendering that resumes after a crash and runs in parallel across processes; 2- and 4-hour 1080p test renders finished with zero duration error and no memory growth
- Built an OpenGL 3.3 compositor (two-pass Lanczos-3) held to a zero-outlier contract of at most 2 levels per channel against the CPU reference: 23/23 scenes pass on an NVIDIA GTX 1650 and on Mesa llvmpipe; 11.7× faster on photographs and 7.7× on transitions at 1080p

AuTrad, quantitative research platform, Indian equities and crypto **September - October 2026**
Creator and sole researcher (code here); paper trading only

- Built a research platform (about 22,000 lines of Python, 222 tests) on point-in-time NSE/BSE data, XBRL filings and the Binance public archive, with Indian trading costs and tax modelled
- Enforced an anti-overfitting protocol: SHA-256-hashed pre-registration before any result (amendments hashed and logged too), an append-only ledger of every trial (114 experiment results, failures included), Holm correction, deflated Sharpe ratio, probability of backtest overfitting and walk-forward tests
- Ran 11 pre-registered research versions (factors, earnings drift, promoter pledges, crypto trend, BTC order flow, market making, funding carry, cash–futures basis, index events); none passed its pre-registered gates, reported openly as "proven alpha: 0"
- When an independent review flagged a look-ahead in my pre-registered basis signal, disclosed it and removed it; froze REV1, the rule that came closest (it later failed the gate at double costs), as a hash-chained paper forward test

Career OS, open-source job-search system (MIT) **September - October 2026**
Creator and sole engineer (code here)

- Built a local system that finds jobs, internships, fellowships and scholarships, scores them, drafts cover letters and autofills applications in a real browser, then stops so the applicant clicks Submit; about 6,800 lines of Python (FastAPI, SQLite, Playwright), 72 API routes
- Wrote job discovery on the public APIs of Greenhouse, Lever, Ashby, SmartRecruiters, Workable and Workday plus four open feeds (112 preset companies), with job-system detection and a headless crawler for custom careers sites
- Ran 13 scheduled agent types; the AI-driven ones use the user's own Claude, ChatGPT or Gemini subscription through their command-line apps, with API keys stripped so they cannot incur API charges. Added usage-limit recovery and Gmail tracking over IMAP that reads only and writes follow-up drafts
- Built a learning area with 16 tracks (78 modules), an AI tutor and tests whose Python coding problems run locally under a time limit

Salesforce AI Agent, open-source agent runtime (MIT) **2026**
Creator and sole engineer (code here)

- Built an agent runtime (Python, FastAPI, async SQLAlchemy, Next.js 15) with 52 tools over the Salesforce REST, Metadata, Tooling and Bulk 2.0 APIs plus Jira, GitHub and Bitbucket; 687 automated tests
- Kept the LLM out of the safety decision: a deterministic risk engine sets the risk tier from environment, object sensitivity, blast radius and production posture, and approvals are bound to a SHA-256 hash of the exact change, an org-state fingerprint and an expiry

Voice Passport, AI Passport Ideathon, Devpost (participant) **August 2026**
Prototype (code here)

- A consent layer for AI voices: a Node.js REST API and creator dashboard that show each request against the creator's policy and, on approval, issue a scoped, time-limited, revocable permission with an audit receipt

Publications

- Kona, R. K. *Where the Time Goes: Profiling, Segmenting and Verifying CPU and GPU Rendering in a Narrated-Video Pipeline.* Preprint, 2026, not yet peer-reviewed (here)
- Kona, R. K. *Blockchain Based Fund Management System.* International Journal of Creative Research Thoughts (IJCRT), 2024
- Kona, R. K. *Blockchain and Its Applications in the Real World.* International Research Journal of Engineering and Technology (IRJET), 2023
- Kona, M. B. and Kona, R. K. *Advancements in Artistic Style Transfer: From Neural Algorithms to Real-Time Adaptation.* IRJET, 2023

Leadership & Activities

CompSoc, University of Edinburgh, Edinburgh, UK **2026 - Present**
Committee Member, Masters Representative

- Elected to the committee of the university's computing society as the voice of Masters students, bringing their views and needs to the committee

UK & Ireland Programming Contest (UKIEPC), University of Edinburgh team **2026**
Competitor

- Competed in the UK and Ireland subregional of the International Collegiate Programming Contest (ICPC), which universities use to choose their teams for the ICPC regional (NWERC); teams from top universities across the UK and Ireland race to solve hard algorithmic problems in five hours

Hackathons & Competitions **2026**
Participant

- Huawei Tech Arena, Huawei UK
- Teachathon, University of Edinburgh
- AI Passport Ideathon, Devpost: built and submitted the Voice Passport prototype (above)
- Global Appathon 2026, MIT App Inventor and App Inventor Foundation (certificate of contribution)

YouTube, technical educator **2025 - Present**
DSA Daily (here) and The Infinite Machine (here)

- Teach data structures, algorithms and computer science from scratch; produce every video myself in OBS Studio

IEEE and university events, India **During BTech**
Presenter and volunteer organiser

- Presented the Web3 Bookstore at IEEE Soft-con Expo and projects at IEEE IoT Expo and the Innovation Fair, JNTU Kakinada; volunteer organiser of ACM events; took part in the Bharat Blockchain Yatra and the Data Science Workshop at IIT Hyderabad

Undergraduate Projects

- **Blockchain-Based Fund Management System**: Solidity smart contracts for decentralised fund allocation with consensus validation and a transaction-transparency interface; published in IJCRT
- **Web3 Bookstore** (IEEE Soft-con Expo): wallet-based authentication; designed the backend APIs and database integration
- **IoT Smart Home Security System**: facial-recognition access control integrated with IoT sensors and an alert-notification backend
- **AIDER** (Innovation Fair, JNTU Kakinada): blockchain-based architecture for securing medical records with decentralised validation of patient data

Certifications

- Salesforce Certified Platform Developer (2024), Platform Administrator (2025) and Platform App Builder (2025); 13 Trailhead superbadges including Apex Specialist (here)
- Microsoft Certified: Azure Developer Associate, DevOps Engineer Expert and Azure AI Engineer Associate (2023)
- Accenture Agentic AI badge (2025) · AWS Academy Graduate: Cloud Architecting and Cloud Foundations (2023)
- UC San Diego: Object-Oriented Java Programming, Data Structures and Beyond (2022) · Meta: Front-End and Back-End Development · DeepLearning.AI: AI For Everyone

Skills & Interests

Languages: Python | Java | TypeScript | JavaScript | SQL | Apex | Solidity
Systems: FastAPI | Node.js | REST/SOAP APIs | SQLAlchemy | PostgreSQL/TimescaleDB | SQLite | Redis | Docker | OAuth 2.0/OIDC | Playwright | multiprocessing | profiling
Data, ML and graphics: pandas | NumPy | SciPy | statistical testing (Holm, deflated Sharpe, PBO) | LLM agents and tool use | MCP | OpenGL | FFmpeg/x264
Platforms: Salesforce (Apex, LWC, SOQL/SOSL, Flow, Metadata/Tooling/Bulk APIs) | AWS (EC2, S3) | Azure and Azure DevOps | Cloudflare | Git and CI/CD | React | Next.js
Honours: Client Recognition, Accenture (2026) | Cheer Award, Accenture (2025) | Ranked 2nd of 66, BTech (2024) | Third Prize, 24-Hour Design Venture Challenge (2022) | Gold Medal, SOF Mathematics Olympiad (2019)
Interests: teaching computer science on YouTube | student representation | competitive programming | hackathons

References: available on request